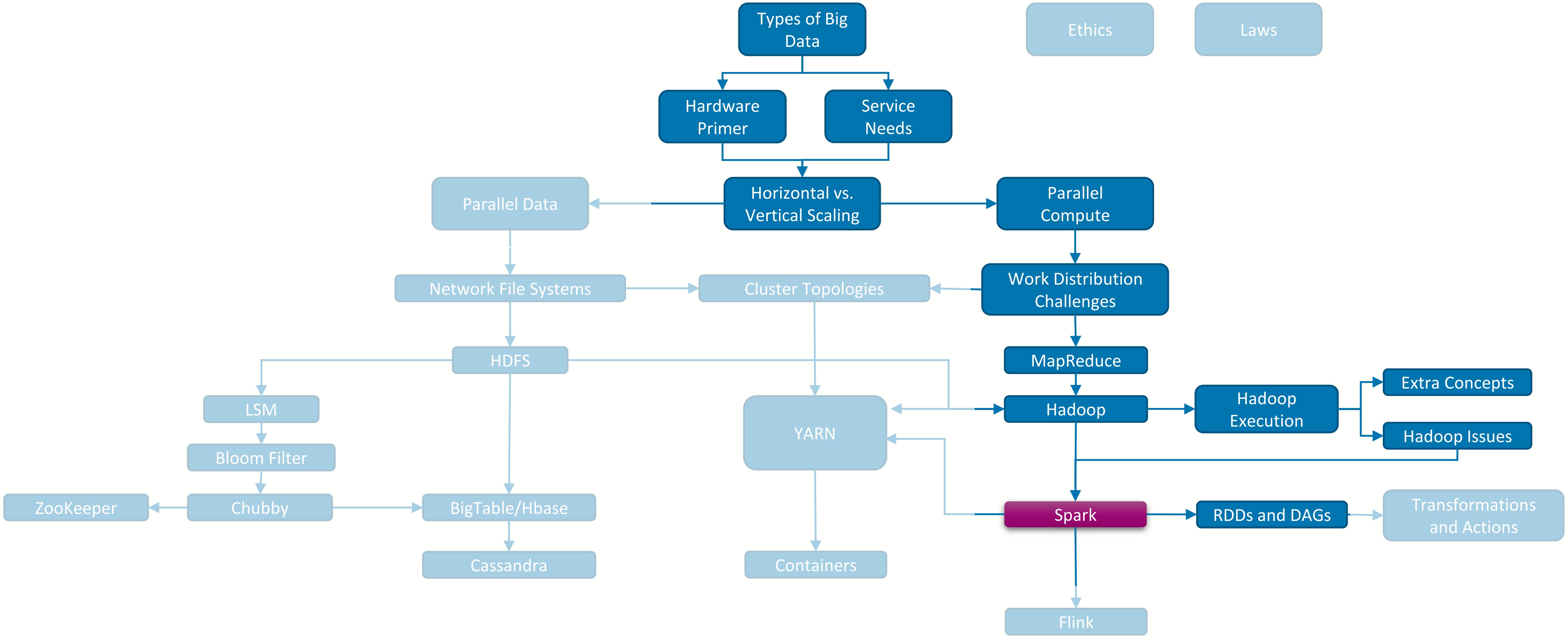
University of Glasgow

W3-05: DAGs

COMPSCI4064 (H) and COMPSCI5088 (M)

Dr. Richard McCreadie

Where are we?


Types of Big Data
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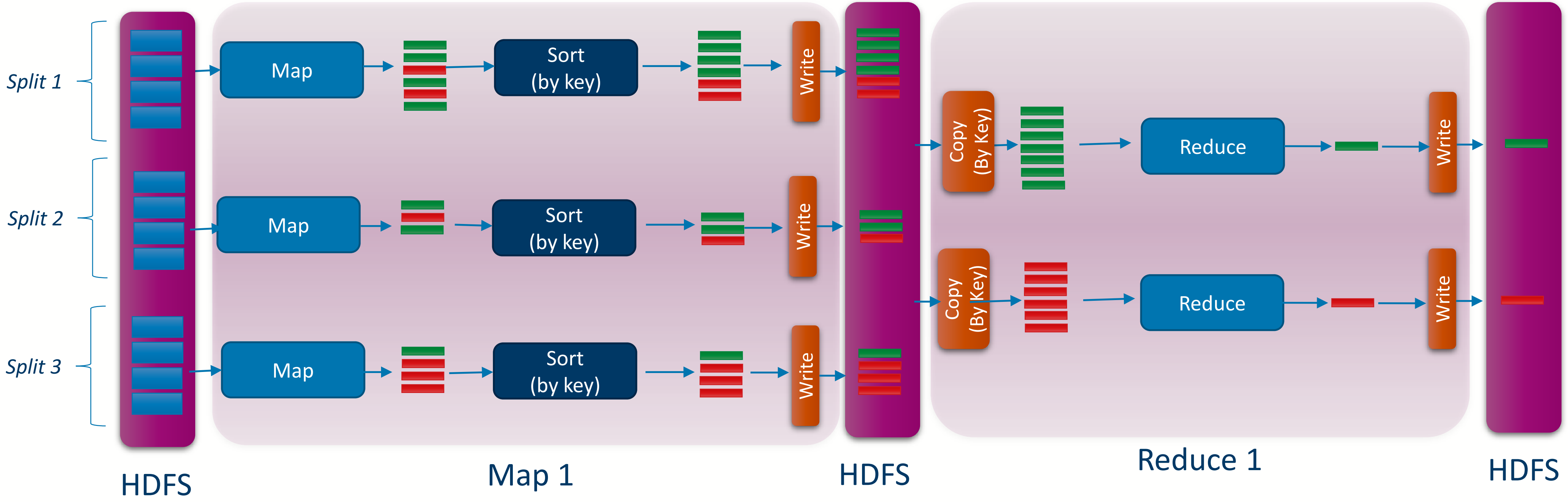
Directed Acyclic Execution Graph (DAG)

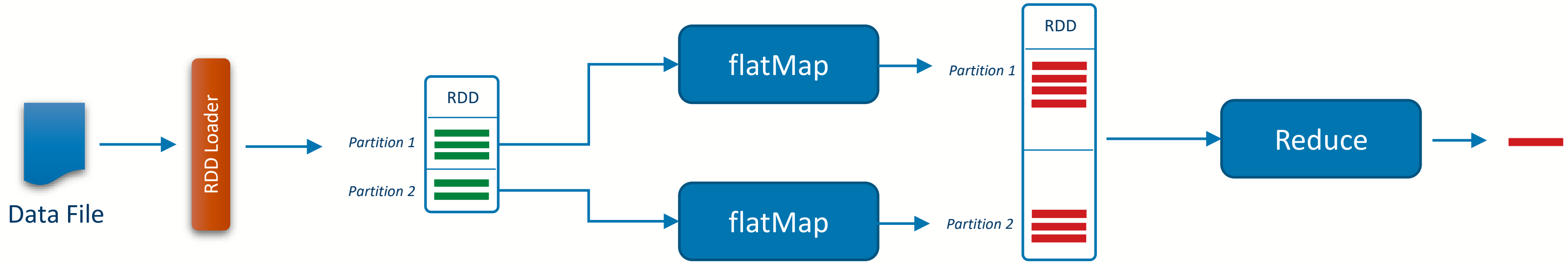
- In Hadoop, the structure of an Job was static…



- …but in Spark, we can chain transformations together

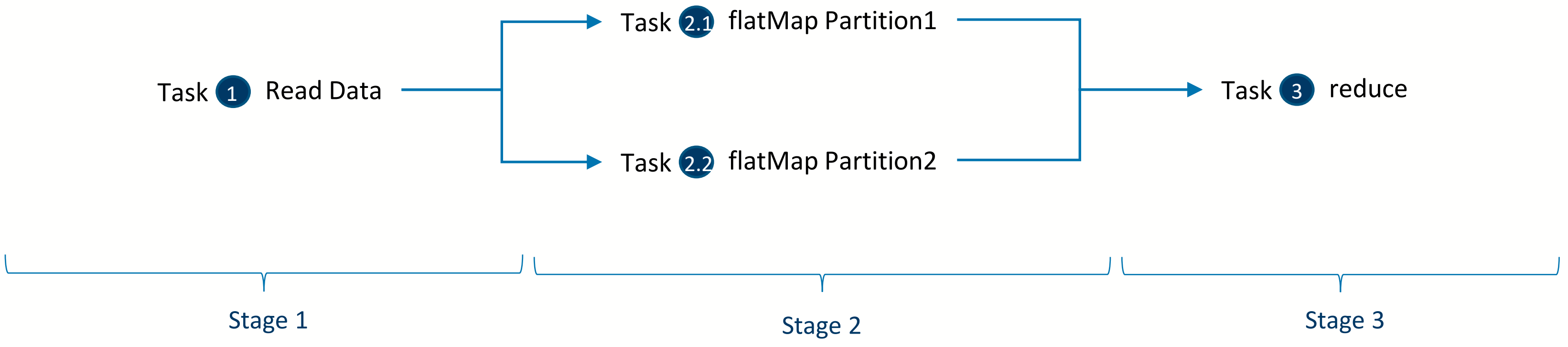
Directed Acyclic Execution Graph (DAG)

- For this reason, in Spark when a Job is submitted the tasks within that Job are compiled into a structure known as a Directed Acyclic Graph (DAG)
  - The DAG represents the order in which tasks need to be executed



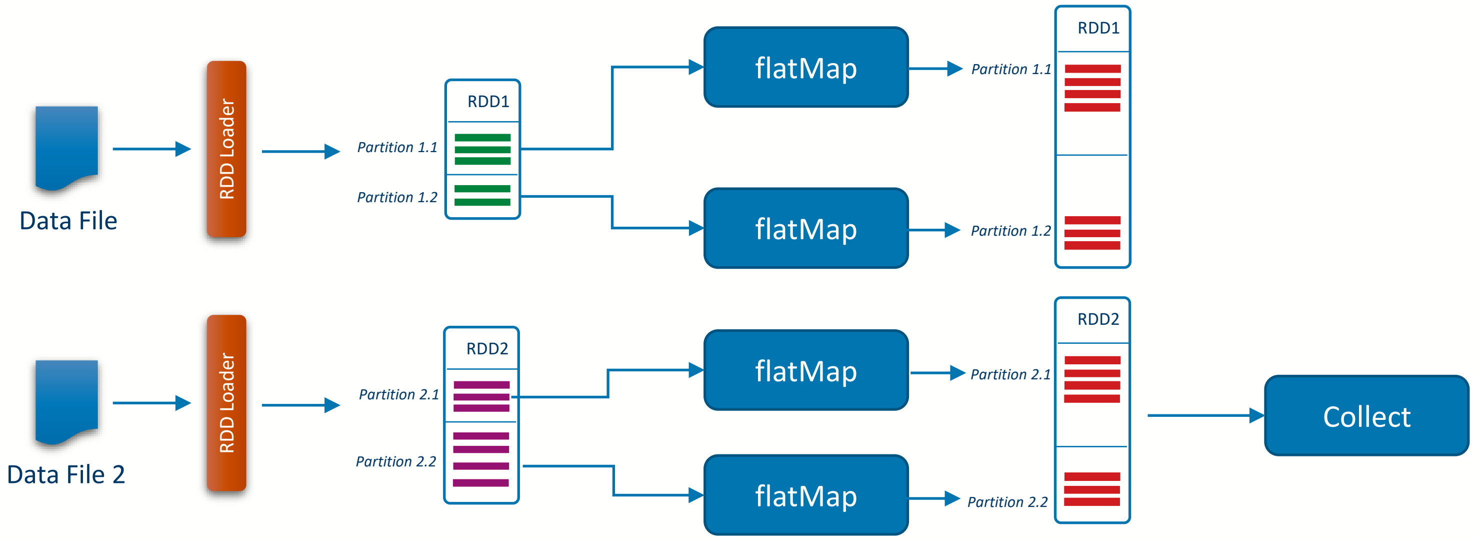
Directed Acyclic Execution Graph (DAG)

- For this reason, in Spark when a Job is submitted the tasks within that Job are compiled into a structure known as a Directed Acyclic Graph (DAG)
  - The DAG represents the order in which tasks need to be executed
  - The Spark scheduler reads the DAG and uses it to identify what tasks can be run in parallel
  - In this way, a Job is divided into Stages, where one stage needs to be completed before the next can begin (because it needs an RDD from a previous stage)



Lazy Execution

- Spark uses what is known as a Lazy Execution Model
  - What this means is that Spark will not schedule a task until its output RDD is needed
  - Actions define the desired output by the user, using the DAG the scheduler can reverse-engineer the steps needed to create that output



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- Spark uses what is known as a Lazy Execution Model
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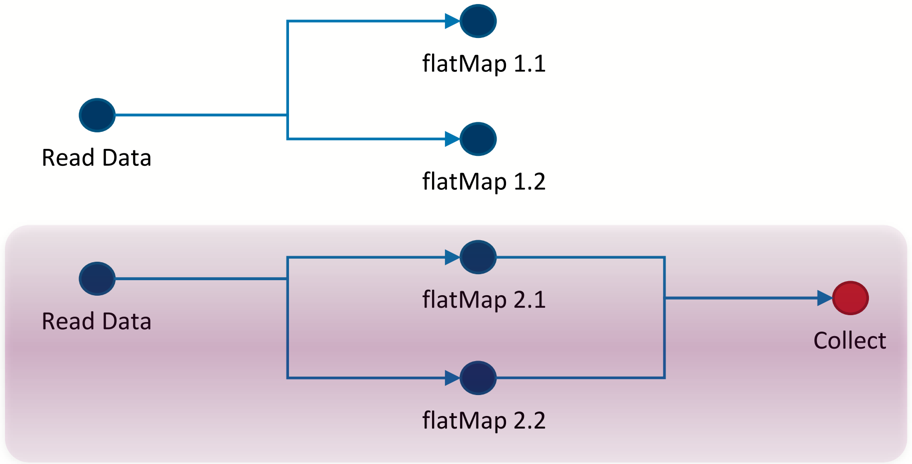
  - Actions define the desired output by the user, using the DAG the scheduler can reverse-engineer the steps needed to create that output
    - Will execute only the minimal tasks needed



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